

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
7th Semester of B. Pharm. Examination
University Theory Examination November/December 2016
PH405 Dosage Form Design

Date: 26/11/2016 , Saturday Time: 10:00 a.m to 1:00 p.m Maximum Marks: 80

Instructions:

- 1. Figures to right indicate marks.**
- 2. Section wise separate answer book to be used.**
- 3. Draw neat sketches wherever necessary.**

SECTION- I

Q 1 Attempt any TWO of the following

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|---|--|----|
| A | Enumerate the physical properties affecting preformulation studies in designing a dosage form and discuss particle size and solubility in details giving suitable illustrations. | 10 |
| B | Define polymorphism and pseudo-polymorphism. Explain the significance of polymorphism in preformulation studies. Enlist the methods to identify polymorphism. | 10 |
| C | Discuss drug-excipient compatibility study and methods of its determination. | 10 |

Q 2 Attempt any TWO of the following

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|---|--|----|
| A | Define kinetics. Discuss the order of reaction with respect of stability testing. | 10 |
| B | Describe effect of hydrolysis on stability of formulation product with example and stabilization against the same. | 10 |
| C | Discuss the protocol for stability testing of tablets. | 10 |

SECTION- II

Q 3 Attempt any TWO of the following

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|---|---|----|
| A | Explain in detail theory of dissolution. | 10 |
| B | Dissolution of dosage form will be depend on which situation. | 10 |
| C | Give the difference:- | |
| | 1. First order model Vs Zero order model | 5 |
| | 2. Statical Method Vs Model Independent for dissolution data analysis | 5 |

Q 4 Attempt any TWO of the following

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|----------|--|------------|
| A | 1. Explain the specification of dissolution apparatus I & II with diagram. | 5 |
| | 2. Theobroma oil can use as suppository base-comments | 2.5 |
| | 3. Tween 80 can be used as wetting agents for liquid dosage form-comments | 2.5 |
| B | Explain in detail the possible interaction of excipients with drug. | 10 |
| C | Give the function, working principal and example of following categories of excipients | 10 |
| | 1. Binder | |
| | 2. Plastizier | |
| | 3. Co solvent | |
| | 4. Chelating agents | |
| | 5. Gelling agent | |
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